

1. Introduction to Farming Systems

- **Definition:** A farming system refers to all the different methods used to produce agricultural products (including crops and animals).
- **Influencing Factors:** The method adopted by a farmer depends on land availability, type of crops and animals, land fertility, skills of the farmer, and availability of funds.
- **Common Systems in Ghana:** Shifting cultivation, land rotation, organic farming, mixed farming, mixed cropping, and crop rotation.

2. Mixed Farming

- **Definition:** A system that involves the growing of crops and the raising of farm animals on the same piece of land at the same time.
- **Components:**
 - * **Crops:** Usually arable crops (cultivated on open fields for one year or a growing season) such as cereals (maize, rice, wheat, millet, sorghum) and legumes (groundnut, cowpea, soybeans, pigeon pea).
 - **Animals:** Ruminants (cattle, sheep, goats) as well as poultry, pigs, rabbits, and grasscutters.
- **Animal Care Systems:**
 - *Extensive system:* Animals roam about the farm in search of their own food.
 - *Intensive system (Cage system):* Animals are kept in cages or pens and fed by the farmer.

Advantages & Disadvantages of Mixed Farming

Advantages	Disadvantages
Feed from Crop Residues: Waste plant materials supply nutrients for animals.	Crop Destruction: Animals may feed on and destroy crops while moving around.
Organic Fertilizer: Animal dung/droppings enrich and fertilize the soil.	Divided Attention: The farmer must split focus between crops and animals.
Reduced Production Costs: Free availability of feed and manure cuts input expenses.	Labour Intensive: Requires a lot of work, often needing hired farmhands.
Farm Power / Labour: Animals like cattle can plough land and cart produce.	High Skill Requirement: Requires extensive knowledge to prevent total failure.
Continuous Income: Multiple sources of income throughout the year.	Year-Round Work: Activities take place continuously, offering no downtime.

3. Mixed Cropping

- **Definition:** A method where a farmer grows two or more crops on the same piece of land at the same time, without any definite row or arrangement.
- **Crop Types:** Cereals, legumes, vegetables, roots, and tuber crops.

Advantages & Disadvantages of Mixed Cropping

Advantages	Disadvantages
Adequate Nutrient Use: Different root depths absorb nutrients at varying levels.	No Mechanization: It is impossible to use machines/equipment due to random planting.
Continuous Harvest: Different crops mature and are harvested at different times.	Expensive Inputs: Different crops may require different, specialized fertilizers.

Advantages	Disadvantages
Pest/Disease Reduction: Difficult for pests and diseases to spread quickly across mixed species.	Time-Consuming: Monitoring multiple types of crops demands significant time.
Improved Fertility: Inclusion of legumes helps fix nitrogen into the soil.	Harvesting Difficulties: Mixed maturity times make harvesting complicated.
Security Against Failure: If one crop fails due to pests, others survive.	Resource Competition: Crops compete heavily for nutrients, water, light, and space.

4. Inter-Cropping

- **Definition:** A system where two or more crops are cultivated on the same piece of land by planting the early-maturing crop *before* the late-maturing crops.
- **Examples:** Sowing maize seeds first, followed shortly after germination by planting cassava sticks. Also, planting young tree crops (like cocoa seedlings) alongside fast-growing plantation crops (like plantain) to provide shade.

Advantages & Disadvantages of Inter-Cropping

Advantages	Disadvantages
Increased Productivity: More than one food crop is harvested from a single unit area of land.	Nutrient Competition: Crops compete for the same soil nutrients at the same place.
Easy Fertilizer Application: The well-defined nature of the farm allows for targeted application.	High Input Demand: Requires more water and fertilizer to ensure all crops develop well.
Easy Pest & Disease Control: Managed spacing makes pest control straightforward.	Mechanization Limits: Specialized machines cannot easily harvest two different crops at once.
Easy Harvesting: Different crops can be harvested easily from the same field.	

5. Crop Rotation

- **Definition:** Growing different types of crops on the same piece of land in a **definite, planned order** over successive growing seasons.
- **Key Principles:**
 1. Must include leguminous crops (e.g., cowpea, groundnut, soya bean).
 2. All crops should ideally mature within one year/growing season.
 3. Shallow-rooted crops should follow deep-rooted crops (e.g., maize following yam).
 4. Crops of the same family or species must not follow one another.
 5. Crops attacked by the same pests and diseases must not follow each other in succession.

Role of Legumes in Crop Rotation

- Leguminous crops (like cowpea) have **nitrogen-fixing bacteria** living in their root nodules.
- These bacteria fix atmospheric nitrogen, enriching the soil with nitrates, which maintains soil fertility for subsequent crops (like cereals).

Advantages & Disadvantages of Crop Rotation

Advantages	Disadvantages
Pest and Pathogen Control: Breaks the lifecycle of pests since the same host crop isn't replanted.	Large Land Requirement: Requires a substantial plot of land to cycle through 4 distinct crops.

Advantages	Disadvantages
Efficient Nutrient Use: Prevents soil nutrients from being overused by a single crop type.	High Knowledge Requirement: Inadequate knowledge of crop behaviors can cause total farm failure.
Increased Yield: Harvesting different crops at different times maximizes total land yield.	
Soil Improvements: Enhances fertility and actively helps to control soil erosion.	

6. System Sustainability & Component Interdependence

All components of a farming system—**land, crops, and animals**—play critical roles in supporting and sustaining one another:

- **How Land Supports Crops & Animals:**
 - Contains essential nutrients and water required for plant growth.
 - Provides physical anchorage for plant roots to stay upright.
 - Yields vegetation (crops/grass) for animal feed and hosts insects that serve as protein for poultry.
- **How Animals Support Land & Crops:**
 - Animal droppings (faeces and urine) serve as **organic manure**, enriching soil fertility and improving crop yields.
- **How Crops Support Land & Animals:**
 - **Soil Structure:** Crop roots bind loose soil particles together, preventing erosion caused by fast-running water.
 - **Shade:** Tall, fast-maturing plants protect young, delicate tree seedlings and animals from harsh sunlight.
 - **Mulch:** Post-harvest crop residues left on the soil surface serve as mulch, which maintains soil moisture, suppresses weeds, and improves fertility for the next season.